

What undergraduate students need to know and actually know about generative AI

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ABSTRACT

In November 2022, the release of ChatGPT sparked widespread adoption of generative AI chatbots among students, yet little is known about what undergraduate students actually understand about these tools or how accurately they perceive their capabilities. To address this gap, we propose a theoretically grounded framework for Generative AI (GenAI) literacy that integrates three forms of conceptual knowledge about large language models (LLMs)—foundations, capabilities, and limitations, and societal impact—with students' perceptions and folk theories of AI chatbots. We then developed and validated a GenAI literacy survey, including multiple-choice knowledge items and perception items, using expert review and item response theory (IRT) modeling. To examine GenAI literacy across different educational contexts, we conducted two complementary studies: Study 1 surveyed students enrolled in courses at a large public R1 university, and Study 2 surveyed a national sample of U.S. undergraduate students recruited via Prolific. Across both samples, approximately 60% of students reported using AI chatbots weekly or daily. However, many students overestimated chatbots' capabilities, particularly on tasks involving reasoning and counting, and often anthropomorphized or treated chatbots as search engines. Knowledge scores were higher among computer science students at the R1 university and frequent chatbot users, while perception accuracy varied by group. Critically, greater conceptual knowledge was associated with less overestimation of chatbot abilities, suggesting that knowledge supports more calibrated and responsible use. This work introduces a validated framework and instrument for assessing GenAI literacy and highlights the need for AI literacy initiatives that move beyond tool usage to address misconceptions, beliefs, and responsible engagement with generative AI in higher education.

1. Introduction

In November 2022, OpenAI released ChatGPT, a general-purpose generative artificial intelligence chatbot powered by large language models (LLMs) trained on extensive text data from the web, books, and academic papers (Hötte et al., 2022). Since the introduction of LLMs, educational researchers, technologists, practitioners, and policymakers have sought to address both the potential benefits and critical challenges posed by generative AI in education. These challenges include concerns such as cheating and academic dishonesty, while the opportunities focus on improving educational practices. For instance, these models have been applied in Socratic tutoring (e.g., Khan Academy's Khanmigo), writing assistance (Shi et al., 2022), teacher assistant training (Markel et al., 2023), and essay scoring (Tate et al., 2024). Additionally, new educational technologies have emerged that enable teachers to create customized chatbots for their students (e.g., Playlab.ai). Meanwhile,

several reports (Baek et al., 2024; Digital Education Council, 2024; Hopelab Common Sense Media & CenterFor Digital Thriving, 2024) indicate widespread use of these models by high school and college students in the United States. Despite known limitations in these models, such as cultural bias (Atari et al., 2023) and hallucinations (Rawte et al., 2023), the adoption of LLMs by students appears inevitable.

While much work has focused on creating tools and environments to enhance students' learning experiences using these models, there is limited research on what students know about these models and how they perceive LLM-based AI chatbots. Assessing students' knowledge about AI more generally is a key focus within AI literacy. Researchers have developed tests (Hornberger et al., 2023; Yau et al., 2022) and pedagogical practices (Kong et al., 2024; Ng et al., 2021; Williams et al., 2023) to evaluate and enhance students' understanding of AI. Given that AI encompasses a wide range of algorithms developed over the past six decades, education researchers approach AI literacy from various

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perspectives and contents, including heuristic algorithms, machine learning, and AI ethics (Ng et al., 2021). However, since the introduction of generative AI tools and their adoption by students, there have been limited efforts to explore students' knowledge of these tools and their perceptions of them, and the relationship between these two. For instance, researchers have looked into students' use of these tools for cheating (Lee et al., 2024) and their attitudes toward and adoption of ChatGPT (Abdalla et al., 2024; Chang et al., 2024); however, there has been little focus on measuring students' knowledge of generative AI or their beliefs about these models' capabilities.

GenAI chatbots, particularly ChatGPT, have gained worldwide attention, with ChatGPT reaching one million users faster than any other technological product in history (Gordon, 2023). An essential aspect of the interaction between students and generative AI is students' ways of perceiving genAI's capabilities, limitations, and fundamentals. For instance, students might overestimate its abilities and consequently over-rely on it (Klingbeil et al., 2024), or they might underestimate its capabilities and incorrectly assume they cannot use it for some tasks. More importantly, the beliefs students develop shape the ways they use AI (Williams et al., 2018).

In this work, we propose a novel framework for GenAI literacy, built upon research in AI literacy and folk theory, and we develop a new GenAI literacy survey to measure students' knowledge about large language models and their perceptions of GenAI chatbots' capabilities in different tasks. In two separate studies, we gathered data from 568 undergraduate students to validate the test and measure students' knowledge and perceptions of GenAI. We find that while about 60% of the students report using AI chatbots¹ for their academic tasks daily or weekly, they tend to overestimate these chatbots' capabilities. However, we find that conceptual knowledge about GenAI relates to the way students perceive its capabilities; when students have more conceptual knowledge about GenAI, they tend to have more realistic views of these models' capabilities and shortcomings. These results underscore the importance of enhancing students' GenAI literacy to shape their utilization of these tools in a more informed way. Understanding these dynamics is crucial for developing effective educational strategies and policies that prepare students to critically and responsibly engage with AI technologies as digital citizens (Touretzky & Gardner-McCune, 2022).

Our findings have implications for educational practices, policy development, and the future of AI integration in education. By understanding and addressing students' misconceptions and knowledge gaps, educators and policymakers can create more effective AI literacy programs, ensuring students are better prepared to engage with AI technologies critically and responsibly. Moreover, this research contributes to the broader discourse on AI ethics and societal impact, highlighting the need for informed public engagement with AI advancements.

2. Theoretical framework

2.1. Different roots: from computational to GenAI literacy

Researchers have posited that AI literacy encompasses— and is rooted in — various types of literacies, such as computational, digital, and media literacy (diSessa, 2000; Logan, 2024; Long & Magerko, 2020).

The concept of computational literacy originates in the constructionist learning theories of Papert (1980) and diSessa (2000). This perspective advocates for programming and computational expression

not merely as technical skills, but as a new form of literacy for developing, articulating, and sharing "powerful ideas." The focus is on active creation and a deep, systemic understanding that emerges from building computational artifacts. While this is a valuable educational paradigm, in this work, we do not view GenAI literacy from the computational literacy lens; In other words, in this study, we are less concerned with what students *do with GenAI*² and more concerned with what they *know and believe about GenAI*. However, as we will discuss later, there are some similarities between our approach and diSessa's in computational literacy, as "mental models" of learners are one of the pillars of computational literacy, as well as one of the main components of our framework.

More direct theoretical precedents arise from (traditional) AI literacy, a field that evolved from digital and media literacy studies. Foundational work by Long and Magerko (2020) defined AI literacy as a set of core competencies (e.g., recognize AI, understand data, think critically) needed to navigate a world infused with often-invisible AI systems like search algorithms and recommendation engines or narrower-use ones such as Alexa. These frameworks are crucial for helping users understand and evaluate the AI they already encounter. However, the introduction of LLMs represents a transformative shift. Unlike earlier AI built for specific, narrow tasks, GenAI chatbots are flexible, open-ended, and generate human-like text, creating a new and explicit interaction paradigm. We argue that this requires a specific GenAI literacy that extends beyond traditional frameworks to address the unique affordances and challenges of generative models.

Within this context, we treat GenAI literacy as a multifaceted construct. In our framework, we define "Conceptual Knowledge" not as a synonym for literacy itself, but as one of its two critical components. As detailed in Section 2.3, we argue that robust GenAI literacy is comprised of both (1) factual conceptual knowledge (i.e., what students should know about LLM foundations, capabilities, and societal impact) and (2) Folk Theories (i.e., what students actually believe, including their beliefs and misconceptions about these tools) (DeVito, 2021).

2.2. Contemporary AI literacy frameworks

Many contemporary AI literacy frameworks are "competency-based," defining what learners should be able to do with AI. These frameworks target specific groups, such as K-12 educators (Miao & Cukurova, 2024), or specific goals, like problem-solving (Kong et al., 2024) or developmental pathways for the workforce (Chee et al., 2024). Their focus is often on AI foundations, ethics, and applications.

For instance, UNESCO's 2024 AI Competency Framework for Teachers (Miao & Cukurova, 2024) tailors AI competencies to educators and defines fifteen competencies across five distinct aspects—"human-centred mindset, ethics of AI, AI foundations and applications, AI pedagogy, and AI for professional learning" (p. 21). Meanwhile, Kong et al. (2024) come up with four different dimensions of AI literacy for problem-solving: "cognitive (conceptual understanding of AI), meta-cognitive (ability to solve problems with AI), affective (psychological readiness to apply AI), and social (ethical considerations around AI-based solutions)" (p. 8). On the other hand, Chee et al. (2024) expand their target learners to include K-16 students, workforce, and general public and propose a competency framework comprising eight core competencies and eighteen sub-competencies, and illustrate how these vary by learner group with an implied developmental pathway from K-12 through higher education to the workforce. Finally, Ng et al.

¹ Although AI chatbots existed prior to the recent wave of LLM-based chatbots, in this article we use the terms "GenAI chatbot" and "AI chatbot" interchangeably. Both terms refer specifically to the LLM-based generative chatbots that have gained global attention following the introduction of ChatGPT and other prominent GenAI models.

² We acknowledge that LLMs can provide valuable opportunities for students to explore powerful ideas many domains such as language—for example, by training small LLMs using datasets of their choosing—this represents a narrower educational use case compared to the mainstream applications offered by GenAI chatbots (For further discussion, see Kafai & Morales-Navarro (2024) and Rismanchian and Doroudi (2025)).

(2021) review how educators define and teach AI literacy, examine their pedagogical practices and content, and categorize these approaches using Bloom's taxonomy (Bloom, 1956) to provide an overview of existing pedagogies on AI literacy.

Long and Magerko's (2020) conceptual framework—perhaps the most prominent framework³—identified five themes that Association for Computing Machinery (ACM) researchers perceive as AI literacy, with seventeen different competencies associated with the themes that are needed by students to not only avoid misconceptions about AI but also prepare for a future where AI is ubiquitous. This framework has become one of the most widely adopted approaches for framing research on AI literacy and addressing students' knowledge needs. While this work and that of others have served as great ways to think about AI literacy, there is a need to update our conceptualizations about AI literacy because the forms of artificial intelligence that students encounter have gone beyond search engines or voice assistants. Nowadays, students can use these models to seek solutions for their homework, request advice about their personal decisions, etc. Hence, our framework adapts important aspects of previous frameworks to address emerging needs with the advent of generative AI and students' knowledge about these tools.

The introduction of LLMs represents a transformative change in how artificial intelligence is accessed and used by everyday individuals. Unlike earlier AI applications, such as Alexa, which were built for a set of specific tasks (e.g., to respond to a specific set of voice commands), LLM-based chatbots are designed to support more flexible and open-ended interactions. This shift has generally made AI more accessible and user-friendly, allowing a much wider audience to explore the capabilities of generative AI. These models are based on advanced architectures that can demonstrate better context awareness and a more sophisticated ability to generate human-like responses in various real-world tasks. This architectural advancement distinguishes LLMs from previous AI technologies by offering a more versatile and dynamic interaction experience. As such, users might naturally resort to one GenAI chatbot for a variety of their tasks instead of choosing specialized AI tools for different problem-solving tasks (Chee et al., 2024).

While these competency models are foundational, some were designed before the widespread public adoption of GenAI chatbots and thus do not adequately center the unique human-GenAI interaction paradigm. Moreover, they often overlook the critical role of users' pre-existing beliefs, intuitions, and "folk theories", which profoundly shape how users interact with, trust, and misunderstand these new tools. This aligns with Logan's (2024) critique that simple competency models "may not be sufficient for understanding, analyzing, and responding" to the complex systems of contemporary GenAI. This disconnect, the need to integrate formal knowledge with informal user beliefs, creates a conceptual gap that our framework is designed to address.

2.3. Our framework

One of the key characteristics of our theoretical framework that sets it apart from the competency-based frameworks discussed above is the recognition of users' and students' beliefs to provide a holistic view of learners' understandings of GenAI. While we tailor important existing components of AI literacy in competency-based frameworks (e.g., Long & Magerko, 2020), cognitive dimensions (Kong et al., 2024), etc.) to measure and address students' knowledge about these models in different domains such as foundations, ethics, and emergent capabilities, we go beyond competency-based models to recognize the existing body of knowledge and beliefs about these models among undergraduate students.

Although students need not—and often do not—fully grasp the technical details of the inner workings of GenAI chatbots, they

frequently develop their own "folk theories" about them—intuitive, informal explanations about the outcomes, effects, or consequences of these technologies. These folk theories of technology guide users' reactions to and interactions with technology (DeVito et al., 2017) in general, and GenAI chatbots in particular. The concept of folk theories is well-established in different academic fields, such as psychology (see Stich & Ravenscroft, 1994; Schlicht, 2023 for an overview), philosophy (Nichols, 2004), and human-computer interaction. For instance, folk theories of technology have been applied to understanding various technological phenomena, such as social media, voice assistants (e.g., Alexa), and other digital tools (DeVito et al., 2017; Druga et al., 2017; Eslami et al., 2015).

Our GenAI literacy not only tries to examine students' conceptual knowledge about generative AI but also aims to understand the way they perceive these models, so it can facilitate future AI literacy efforts and curricula to be responsive to common presumptions, misconceptions, and knowledge that students possess.

As such, our framework consists of two intertwined components: Knowledge and Perception of Generative AI. The Knowledge component is composed of three constructs: "What is generative AI?", "Emergent capabilities and open problems in generative AI," and "GenAI and societal impact." The Perception component is composed of a single construct: "Perceptions of AI chatbots." While we have separated these components out, we note that they are actually intertwined insofar as students' knowledge of AI informs their perceptions of AI and vice versa. Fig. 1 provides a depiction of our framework, and Table 1 shows how the constructs in our framework build upon prior work in AI literacy and folk theory.

In the knowledge component, for the first construct, we focus on measuring participants' ability to distinguish between different types of intelligent systems (e.g., human-level understanding, search engines, and basic algorithmic functions) and their ability to qualitatively explain how LLMs work. The second construct aims to assess participants' awareness of the limitations and strengths of this technology. Finally, the third construct targets the ethical and societal challenges associated with these models.

2.4. The present study

Based on the theoretical framework outlined above, our study aims to operationalize and test the intertwined relationship between students' conceptual knowledge and their perceptions of GenAI. We seek to map the current landscape of GenAI literacy among undergraduates and identify key factors that correlate with both knowledge and perception.

To this end, we posit the following hypotheses:

First, the role of experiential learning (i.e., frequent use) is complex. While use can lead to familiarity, it does not guarantee conceptual understanding and may even reinforce folk theories. We aim to explore this relationship, hypothesizing that frequent interaction provides more opportunities for both learning and misconceptions.

H1. Frequency of chatbot use will be associated with both Conceptual Knowledge scores and students' overestimation.

Second, our framework is predicated on the idea that students' informal "folk theories" are a critical component of their literacy. Given the human-like and black-box nature of chatbots, it is likely that these folk theories include significant misconceptions, such as anthropomorphism (Li & Suh, 2021) or overestimation, as suggested by prior work.

H2. Students will demonstrate significant misconceptions about GenAI, particularly in overestimating its capabilities. In other words, when judging these models' capabilities, they tend to overestimate actual capabilities more than they underestimate.

Third, literature on digital and AI literacy suggests that prior educational exposure is a key factor in knowledge acquisition (e.g., Baek et al., 2024; Kong et al., 2024). Furthermore, recent work points to emerging educational inequalities in how students access and

³ As of this writing, it has received more than 1900 citations per Google Scholar metrics.

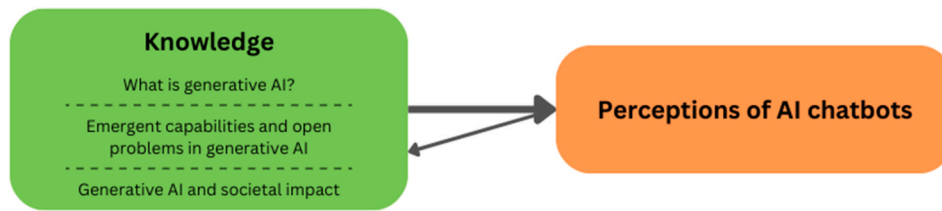


Fig. 1. Our Theoretical Framework: Knowledge and perceptions of GenAI are intertwined and influence one another.

Table 1

The interrelationships between our framework, Long and Magerko (2020), and folk theory (DeVito et al., 2017).

Theoretical Frameworks	Themes	Our Framework	keywords
How do researchers define AI literacy? What are the essential competencies that students need regarding AI? Long and Magerko (2020)	What is AI?	What is generative AI?	Emergence vs. by training, dataset size, RLHF, probabilistic next token prediction, prompting
	What can AI do?	Emergent capabilities and open problems in generative AI	Hallucinations, unlearning, logical reasoning, prompt injection,
	How does AI work?		
	How should AI be used?	GenAI and societal impact	Guardrails, jailbreaking, RLHF, Bias, AI alignment
Folk theory (DeVito et al., 2017)	How do people perceive AI? What are people's informal theories about technology?	Perceptions of GenAI chatbots	Search engines, human-like reasoning, databases

understand AI (Yu et al., 2024). We therefore expect that students in technical fields, who have more formal exposure to computational concepts and are usually from a higher socioeconomic status (Koch et al., 2024), will have a more developed conceptual understanding.

H3. Students with backgrounds in computer science will demonstrate significantly higher conceptual knowledge scores than students in non-CS fields.

Finally, and most central to our proposed framework, is the intertwined nature of knowledge and perception. We posit that factual knowledge is a key mechanism for calibrating user perceptions. Therefore, students who understand how LLMs work (conceptual knowledge) should be better at identifying what they can and cannot do (folk theories).

H4. Higher conceptual knowledge scores will be associated with lower overestimation of these models' capabilities.

3. Methods

3.1. Survey design

We designed a survey on generative AI literacy with a focus on large language models, in which there are two major types of questions: 1) *knowledge questions*, including 14 questions with four options (except for two questions with three options) where only one is correct and is designed to capture what general knowledge about generative AI students possess and 2) *perception questions*: including 17 five-point Likert

items which are designed to investigate how participants estimate strengths and limitations in prevalent AI chatbots performing specific tasks and to what extent their perceptions are close to chatbots' actual capabilities. The knowledge questions and perception questions naturally relate to the Knowledge and Perception components of our framework, respectively.

To ensure the validity of our questions, we held a session with an expert in the field of natural language processing and employed a think-aloud technique to gain feedback on the correctness and accuracy of our questions and their options. Based on the session, we updated two of our questions and removed one. We further pilot-tested our knowledge questions with a group of undergraduate students and collected their feedback on the readability and wording of the questions, and updated the text of a few of our questions based on it.

3.1.1. Knowledge questions

We designed our knowledge questions based on our theoretical framework. For each of the three constructs in the Knowledge component of our framework (e.g., what is generative AI?), we designed a few questions to assess participants' knowledge in that specific construct. We designed the questions and the distractor options based on recent literature in computer science that includes research that we believe should inform students' AI literacy across the three constructs. Table 2 shows each knowledge question, its corresponding construct in the framework, and related papers in the computer science literature that introduce the associated ideas or issues; the exact questions and answer choices are given in the Appendix. In addition to the types of questions, the distractor options were crafted to reflect common misunderstandings identified in prior literature on AI literacy (e.g., Druga et al., 2017; Eslami et al., 2015), capturing misconceptions people may have about GenAI, such as believing that ChatGPT reasons like humans or that these models simply search the web to generate responses.

3.1.2. Perception questions

We designed 17 different perception questions where we asked participants to assign a likelihood to an AI chatbot coming up with a correct response to a prompt. For instance, one of the questions in the survey is as follows: "How likely do you think an AI chatbot is to answer the following question correctly? 'Where was the current President of the United States born?'" The domain of these prompts spans a wide range of areas, including information retrieval (4 tasks), safety measures (1 task), commonsense questions (3 tasks), math problems (3 tasks), counting tasks (2 tasks), text transformation (1 task), and automatic text generation (3 tasks). The tasks were chosen based on related literature that reported the strengths of these models in text generation tasks (Bubeck et al., 2023) and those that reported specific challenges and limitations, such as mathematical or commonsense problems (Cherian et al., 2023; Mitchell, 2023).

3.2. Study 1 - R1 university

Our first study focuses on two intact classes at a large public R1 university in the United States. In this study, we surveyed (1) computer-science students enrolled in an introductory AI course (which did not cover advanced NLP methods such as LLMs) and (2) psychology and

Table 2

Knowledge questions in our survey and corresponding constructs. Note that the wording of questions in this table differs from what is represented in the survey for the sake of brevity.

Construct	Question	Related Literature
What is generative AI?	How do large language models generate text?	Vaswani et al., 2017; Wei et al., 2022
	How do large language models generate programming code?	Vaswani et al., 2017; Roziere et al., 2023
	What is a prompt?	Wei et al. (2022)
	What is the reasoning procedure in LLMs when they play chess?	Duan et al. (2024)
	What does hallucination or confabulation mean in the domain of LLMs?	Rawte et al., 2023;
	How can we have an LLM forget data?	Eldan and Russinovich (2023)
	What is the reason for an LLM to answer a math problem wrong?	Razeghi et al. (2022)
	What is the reason for an LLM to identify a non-sensitive question as a sensitive question?	Weidinger et al. (2022)
	How do LLMs avoid answering sensitive questions?	Chao et al., 2023; Yong et al., 2023
	Is it possible to have an AI chatbot generate problematic or sensitive text?	Chao et al., 2023; Yong et al., 2023
Emergent capabilities and applications and open problems in generative AI	Do LLMs work well in all different languages?	Robinson et al. (2023)
	Is it possible for public or independent researchers to investigate the data used for training proprietary chatbots such as ChatGPT?	Balocco et al. (2024)
	Can LLMs be biased toward some cultures?	Atari et al. (2023)
	Does an AI chatbot answer a specific sensitive question about the best professions for different genders?	Ghosh & Caliskan (2023)
Generative AI and societal impact		

education sciences students enrolled in a course on cognition and learning. We chose these cohorts because we hypothesized they would differ markedly in both technology use and understanding of CS and AI concepts—specifically, that computer-science majors would engage with AI tools more extensively than education and psychology majors.

Participants were recruited by email and were offered 2% extra credit for their participation. As per IRB requirements, an alternative task was provided for those who did not wish to participate in the research; they could watch a YouTube video on generative AI and write a short summary paragraph to receive the extra credit. This phase of the study was conducted during March 2024. Overall, 257 students were recruited for this study.

3.3. Study 2 - US undergraduate students

In the second study, during May 2024, we used Prolific (Peer et al., 2022) to gather data from a wider range of students throughout the US who may be more representative of higher education students in the US compared to those of R1 University in California. We note that Prolific users may still have their own idiosyncrasies that are not representative of university students (e.g., they may use AI more than the average university student), so by including both types of participants in our research, we can test the hypotheses in our research question in two different sets of populations, to better understand parallel trends and differing ones.

To this end, we gathered data from undergraduate students in the United States with an exclusion criterion of age under 18 or higher than 40. The median time it took each participant to complete the survey was around 15 min, and we compensated each participant at an average rate of \$17.52 per hour. In this phase, 334 participants were recruited. We excluded 23 participants who failed to answer two attention questions correctly, resulting in a final sample of 311 in this study.

3.4. Aggregate demographics

The aggregated sample demographic of our participants comprises 44% computer science students, 13% STEM students except for computer science, and 43% non-STEM students. Regarding participants' gender, 54% are male, 44% are female, and 1% identify themselves as having other gender identities.

Participants' demographics, as well as their reported chatbot usage frequency and the type of chatbots that they use, are reported in Table 3. Among usage statistics, we did not find a significant difference between the reported data from the groups of our participants.

We asked participants what chatbots they used the most, and 68% of participants reported using GPT-3.5, 19% using GPT-4 (the paid version of ChatGPT with higher capabilities), 8% using Gemini/Bard, and the rest using other chatbots. This finding is notable as the differences in capabilities between GPT-3.5 and GPT-4 are considerable (Bubeck et al., 2023), and most of our participants were using free versions that lagged behind the state-of-the-art models.⁴

Furthermore, 17% of our participants reported using AI chatbots daily, 43% weekly, 20% monthly, and the rest 20% rarely or never on average, which shows the widespread usage of these tools among undergraduate students in the US. However, we want to note that as most of the participants are recruited from either a CS class or from Prolific—itsself a technology platform—it seems possible that the adoption rate of AI chatbots in our sample may be higher than for undergraduate students in large.

Finally, the overall proportion of English language learners (ELLs) in our final sample was 8.9%, slightly below the national undergraduate average of 10.6% (National Center for Education Statistics, 2024). Within our subgroups, 18% of R1 university students were ELLs (consistent with California's average, which is 18.9%), whereas only 2% of Prolific participants reported ELL status. This contrast highlights a potential limitation of the Prolific sample in representing the broader population of U.S. undergraduates.

3.5. Analysis

3.5.1. Reliability and measurement invariance

Similar to related work on AI literacy tests (Hornberger et al., 2023), we use Item Response Theory (IRT) models using the *mirt* package in R (Chalmers, 2012) to analyze the knowledge component of our survey and its reliability. In these models, participants' abilities and item characteristics are estimated on a joint scale (Embretson & Reise, 2013). IRT offers several advantages over classical test theory, such as the ability to analyze test data at the item level. The three most prominent IRT models are the Rasch (1-PL) model, the 2-PL model, and the 3-PL model (Embretson & Reise, 2013). These models differ in the number of parameters they estimate for item characteristics. The Rasch model estimates only one parameter for item difficulty. The 2-PL model additionally estimates parameters for item discrimination, and the 3-PL model extends the 2-PL model by adding a guessing parameter (γ), which reflects the likelihood of participants guessing the correct answer, enhancing the model's ability to differentiate between varying levels of participant ability.

⁴ Since then, OpenAI made the GPT-4 model free to users, but at the time that we conducted the survey, only GPT-3.5 was available with no payment.

Table 3

Descriptive demographic data of participants in our survey.

	N	CS majors	Other STEM majors	Non-STEM majors	Male	Female	Never used chatbot	Daily or Weekly usage	Using Chat-GPT	Using GPT-4
R1	257	77.2%	0%	22.8%	62.4%	37.1%	19.8%	66%	64.1%	19.5%
Prolific	311	21.1%	22.1%	56.8%	47.7%	49.2%	21%	55.4%	72.8%	19.5%
Total	568	44.5%	12.9%	42.6%	53.9%	44.2%	20.5%	60.2%	67.9%	19.5%

Before fitting the IRT models for the analysis of knowledge questions, we tested for unidimensionality and local independence assumptions in our data separately for both of the studies and we found that the unidimensionality assumption held true; unidimensionality was measured using a confirmatory factor analysis (CFA) model, resulting in root mean square error of approximation (RMSEA) < 0.03 , standardized root mean squared residuals (SRMR) < 0.05 , $\chi^2/df < 2$. On the other hand, for the combination of knowledge and perception questions, we used Cronbach's alpha statistic to measure the internal consistency of our survey, and we found that $\alpha = 0.82$, suggesting that the survey is highly reliable in measuring the underlying construct.

To examine whether the knowledge test functioned equivalently across the two studies (R1 vs. Prolific), we conducted item-level Differential Item Functioning (DIF) analyses using Wald tests in a multi-group 2PL IRT model (Chalmers, 2012). We tested both uniform DIF (differences in item difficulty/intercepts) and non-uniform DIF (differences in item discrimination/slopes). To control for the false discovery rate across multiple tests, we applied the Benjamini–Hochberg (BH) correction (Benjamini & Hochberg, 1995). As shown in Table 12, no items were flagged for significant DIF (adjusted $p < .05$). This indicates that the items function similarly across groups, supporting measurement invariance and justifying the comparison of ability scores across studies, if needed.

3.5.2. Students' overestimation

Following the excitement that the introduction of ChatGPT brought to students, educators, and technology leaders, many researchers warned about the ethical challenges of these tools as well as the hype around them that might mislead people to overestimate the abilities of these models (Rudolph et al., 2023). Given the circumstances, we hypothesized that university students might tend to overestimate the abilities of current chatbots. We aimed to first answer the question about the proportion of participants overestimating the abilities of the chatbots. To measure if a participant is overestimating a chatbot's specific ability, for each perception question, we evaluated the performance of the most commonly used chatbots—GPT-3.5, GPT-4, and Gemini/Bard—against specific prompts to determine their success rates. Each chatbot was tested five times with the same prompts, and we recorded how often they successfully answered the questions.⁵ The performance of each chatbot on each question was then calculated by averaging the number of successful attempts. To obtain an actual performance score for the chatbots, we averaged the individual performances of the three chatbots, assigning scores on a scale of 1 to 5. These actual performance scores provided a baseline for evaluating participants' estimates of chatbot capabilities. By subtracting the chatbots' average actual ability from the students' assigned ability (using Likert scores), we were able to identify instances of overestimation or underestimation. If the result is larger than 1, then the student is counted as overestimating that specific ability, and if the result is smaller than -1 , counted as underestimating.

A potential question regarding this approach is why we did not compare each participant's answers to the performance of their most-used chatbot. While it seems reasonable to assume that users of different chatbots might have significantly different perceptions, our

data did not support this. We conducted a two-sided t -test to examine whether users of different chatbots had significantly different perceptions, and out of 17 questions, significant differences were found in only 4. Notably, in 3 of those 4 questions, the chatbots had the same performance. Therefore, we chose not to personalize this measure and instead used an aggregated measure of chatbot performance in our analyses.

We further used linear regression models within our studies with R1 and Prolific students, using overestimation percentage as the dependent variable and demographic variables, as well as students' derived abilities from knowledge questions as predictors.

3.5.3. Students' misconceptions

To explore different patterns of beliefs and theories that participants develop about GenAI, we need to delve into their choices in knowledge and perception questions. We employed the K-modes clustering method (Chaturvedi et al., 2001). K-modes clustering is an unsupervised machine learning algorithm tailored to categorical data. Unlike k-means clustering, which relies on calculating the mean, K-modes employs the mode to determine cluster centroids, making it well-suited for non-numeric data. This algorithm iteratively assigns data points to clusters based on the similarity of their categorical attributes, updating the centroids to minimize the total distance within clusters.

After running the elbow method to determine the optimal number of clusters, we found that the elbow point appeared at $k = 8$ for within-cluster simple-matching distance for each cluster. However, considering that eight clusters seemed excessive for our dataset, we decided to manually test for other indices of clustering using NbClust package (Charad et al., 2014) in R. Through this process, we calculated the dissimilarity matrix using Gower method and using the matrix we found that by majority voting of five different indices, $k = 3$ was chosen as the optimal number for our clustering. $k = 3$ had the highest scores in Silhouettes (Rousseeuw, 1987) and Dunn (1974) indices, showing that it is an appropriate number of clusters, balancing interpretability with within-cluster homogeneity. This allowed for a more coherent analysis of participants' patterns.

4. Results

4.1. Measurement

4.1.1. Model selection

To select the final measurement model, we compared the fit of the 1-PL (Rasch), 2-PL, and 3-PL multigroup models. All models were specified as invariant, constraining item parameters to be equal across groups as justified by our prior Differential Item Functioning (DIF) analysis in Section 3.5.1.

Model fit statistics are presented in Table 4. The multigroup Rasch model demonstrated a poor fit to the data (TLI = 0.836, CFI = 0.825, RMSEA = 0.038). The multigroup 2-PL model, which allows item discrimination parameters to vary, offered a substantial improvement and an acceptable fit (TLI = 0.901, CFI = 0.902, RMSEA = 0.029).

However, the multigroup 3-PL model provided the best fit by a clear margin. This model, which accounts for both item discrimination and a guessing parameter, yielded excellent fit indices (TLI = 0.948, CFI = 0.956, RMSEA = 0.021). The theoretical appropriateness of a guessing parameter for a knowledge-based assessment, combined with its

⁵ The tests took place in March 2024.

Table 4

Model fit indices for all Rasch, 2-PL, and 3-PL models. The 3-PL model suggests the best fit.

Model	M2	RMSEA	SRMSR-R1	SRMR-Prolific	TLI	CFI	AIC	BIC
Rasch	349.26	0.04	0.09	0.07	0.84	0.83	9537.57	9607.04
2-PL	266.93	0.03	0.07	0.05	0.90	0.90	9486.39	9616.65
3-PL	190.63	0.02	0.06	0.05	0.95	0.96	9453.32	9705.16

Note. M2 = M-squared goodness-of-fit statistic; RMSEA = root mean square error of approximation; SRMR = standardized root mean squared residuals (reported for both study groups), TLI = Tucker–Lewis index; CFI = comparative fit index; AIC = Akaike information criterion; BIC = Bayesian information criterion.

superior empirical fit, strongly supports its selection. Therefore, the invariant multigroup 3-PL model was adopted as the final model for estimating item parameters and student abilities (θ) for all subsequent analyses.

Item fit statistics for the 3-PL model is shown in Table 5. To assess item-level fit, we examined the signed chi-square ($S-\chi^2$) statistic for each item within each group (Table 5). While the $S-\chi^2$ p-value is a common index, it is known to be overly sensitive to large sample sizes (de Ayala, 2009). Therefore, we also considered the RMSEA. $S-\chi^2$ as an effect size to gauge the magnitude of any misfit (Chalmers, 2012). The results indicated that 12 out of 14 items fit well with data from both studies, and on the other hand, the item 'Professions' showed a significant p-value ($p = .021$) in the Prolific group, but its RMSEA. $S-\chi^2$ of 0.071 suggested an acceptable magnitude of misfit (Chalmers, 2012). Furthermore, the item 'Hallucination' demonstrated poor fit in the R1 group, as evidenced by a significant p-value ($p = .02$) but also with an acceptable RMSEA. $S-\chi^2$ (0.080). Given that only two items showed poor fit (as judged by p-values) but with an acceptable misfit magnitude (as judged by RMSEA. $S-\chi^2$) and the overall model fit was excellent, we retained all items for the final analysis to maintain content validity. However, we acknowledge that the 'Hallucination' and 'Professions' items show misfit with one of the groups, and their parameters should be interpreted with caution.

Finally, using the 3-PL model, we examined the assumption of local independence, which posits that the responses to test items are independent of each other given the participant's ability level. We assessed this assumption using correlations of residuals for the Q3 index across all items. According to Chen and Thissen (1997), local independence is upheld when these correlations are smaller than 0.2. Our analysis confirmed that all correlations fell below this threshold, affirming the assumption of local independence within our test data.

4.2. How do university students perform in our GenAI literacy survey?

Using the ability variable from the multigroup 3-PL model, we can

Table 5

Item fit statistics. 2-PL and 3-PL models show a good fit for all items.

Item	R1				Prolific			
	$S-\chi^2$	df	RMSEA	p	$S-\chi^2$	df	RMSEA	p
TextGen	11.97	6	0.065	0.063	9.11	5	0.05	0.105
CodeGen	8.11	6	0.039	0.23	3.98	4	0.000	0.408
IncorrectMult	11.68	6	0.063	0.07	7.89	5	0.042	0.162
RefuseReq	1.99	7	0.000	0.96	4.45	6	0.000	0.616
PromptDef	13.17	7	0.061	0.068	6.84	5	0.033	0.233
Hallucination	13.27	7	0.082	0.020	3.43	5	0.000	0.634
ForgetData	7.91	8	0.000	0.442	7.9	6	0.031	0.246
Professions	9.49	6	0.05	0.148	13.32	5	0.071	0.021
ChessReasoning	6.58	7	0.000	0.474	2.03	5	0.000	0.846
SensitiveAns	10.4	6	0.056	0.109	3.73	4	0.000	0.443
ReligionQ	6.87	7	0.000	0.442	9.05	6	0.039	0.171
LangPerf	11.24	7	0.051	0.129	11.97	6	0.055	0.063
LLMBias	7.79	6	0.036	0.254	6.81	3	0.062	0.078
LLMData	8.91	7	0.034	0.259	4.19	5	0.000	0.522

Note. $S-\chi^2$ = Orlando–Thissen item-fit statistic; df = degrees of freedom; RMSEA = item-level RMSEA from $S-\chi^2$; p = p-value for $S-\chi^2$. Values in bold indicate poor fit ($p < .05$).

derive participants' knowledge in Generative AI. As Fig. 2 shows, students in the R1 university perform better than students across the US on average. However, a finer-grained look inside our studies shows that in the R1 university, high-achievers are mostly the CS students, while in the Prolific sample, we see no such clear difference in the performance of students from different major types.

4.2.1. R1 students performance

Among the students in the R1 university, perhaps unsurprisingly, students of the CS major performed significantly better (Table 6). As the regression model's results are shown in Table 6, a higher frequency of chatbot usage is associated with higher conceptual knowledge about these models. Furthermore, female students performed significantly better than their peers, and interestingly, users of the paid version of the GPT model performed worse on the test. We found no significant relationship between ELL status and performance on the test (see Table 7).

4.2.2. Prolific students

Among undergraduates across the US, we see a parallel pattern of associations between students' derived abilities and their chatbot usage frequency and students' gender. Similar to Study 1, higher chatbot usage frequency is significantly associated with higher knowledge about GenAI models. Female students performed significantly better on the test. Similar to study 1, we found no association between ELL status and derived abilities among Prolific students.

4.2.3. Aggregated results

As discussed in the subsections above, we conducted separate regression analyses to better understand factors that relate to students' conceptual knowledge of the models. We only found one pattern regarding the H1 and H3 hypotheses of this paper that was replicated in both studies: the association between higher usage and higher conceptual knowledge (H1). We rejected the null hypothesis regarding the null relationship between these two. Furthermore, regarding H3, which claims CS students perform better than their peers in other major types,

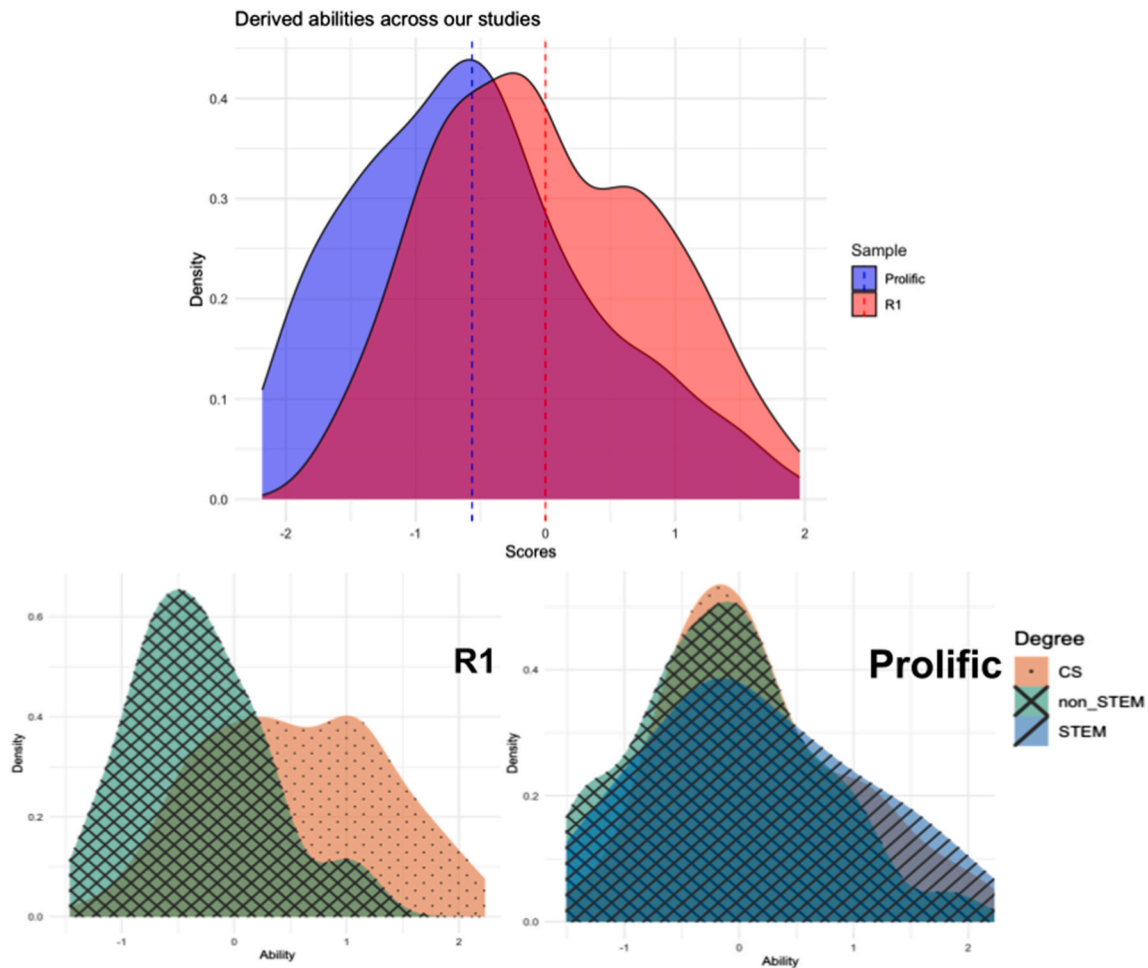


Fig. 2. Distribution of derived abilities across and within our studies, (top) students in R1 university perform better than US students recruited by Prolific on average, (bottom-left) CS students at the R1 university perform better than their peers in Non-STEM majors, (bottom-right) students across the US of different major types perform comparably.

we only found evidence within the R1 university study, where CS students performed significantly better than their peers. This pattern was not replicated in study 2. As we will discuss in the discussion sections, these discrepancies might stem from CS students' more advanced knowledge in AI and computer science concepts, their different socioeconomic status and access to technology, or other important reasons.

4.3. What are students' perceptions of generative AI?

4.3.1. Students overestimate GenAI's capabilities too often

We first hypothesized that most students tend to overestimate more often than underestimate (H2). As Fig. 3 shows, where each student is depicted as a dot on the scatter plot, where x is their overestimation percentage and y is their underestimation percentage, most of the participants lie under the $x = y$ line, which shows that our hypothesis holds true, on aggregate, and both within our subgroups. Aggregatedly, we find that non-STEM students tend to overestimate more compared to other groups (36.9% vs. 31.5% on average, $p < .001$), and interestingly, computer science students in the R1 university tend to underestimate more than other groups (18.01% vs. 11.23% on average, $p = .001$).

One of the common types of questions that students overestimate is the ones that ask about AI chatbots' abilities to count. For instance, one question was asking about the ability of an AI chatbot to come up with five words that each have two "a"s as in "again". Although the question should be simple for humans, it is not the case for LLMs, as they lack the ability to count. Usually, when this question is asked by an AI chatbot,

they come up with words such as "banana", but most of the users (84%) assigned "very likely" for their chatbot to come up with the correct response to the question. On the other hand, we asked if their AI chatbot can write a paragraph that only includes the word "the" seven times; in this case, similarly, students assigned high likeliness for the chatbots to respond successfully to this prompt (45% assigned "very likely") but as our tests show, the models are not accurate in these text generation regimes as they are probabilistic prediction models and they lack the accurate ability of counting.

On the other hand, when the questions were about tasks such as information retrieval about past events (e.g., the winner of Super Bowl 2022), participants were successful in determining the abilities of chatbots. However, users of GPT-3.5 also predicted a high likelihood of their chatbot to answer a question about recent events, such as the winner of Super Bowl 2024 (44% of GPT-3.5 users), while we know that this model did not have internet access and its data was updated until 2023, so it would have not been able to respond to the prompt successfully.

4.3.2. More knowledge reduces overestimation

As previously mentioned, one of the assumptions underlying our work is that increased knowledge about GenAI can enable students to use these tools in a more constructive and informed manner (H4). This awareness includes understanding the strengths and limitations of these models, thereby allowing students to make informed decisions about when to utilize them and when to refrain from relying on them. To

Table 6

Regression in Study 1, results using ability as the dependent variable for R1 students.

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]	<i>sr</i> ²	<i>sr</i> ² 95% CI [LL, UL]	Fit
(Intercept)	0.78**	[0.26, 1.30]			
ChatbotFreq	0.11*	[0.01, 0.21]	0.02	[-0.01, 0.04]	
ChatbotUsed (Baseline: Gemini/Other)					
GPT-4 (paid version)	-0.57*	[-1.01, -0.12]	0.02	[-0.01, 0.05]	
GPT-3.5	-0.31	[-0.70, 0.07]	0.01	[-0.01, 0.03]	
Gender (Baseline: Male)					
Female	0.48**	[0.24, 0.72]	0.05	[0.00, 0.10]	
Other Gender	0.19	[-1.28, 1.65]	0.00	[-0.00, 0.00]	
TimeTaken	0.00	[-0.00, 0.00]	0.00	[-0.00, 0.01]	
Non-native English Speaker	-0.22	[-0.47, 0.03]	0.01	[-0.01, 0.03]	
Non-STEM	-0.55**	[-0.83, -0.27]	0.05	[0.00, 0.10]	
					R ² = 0.276** 95% CI [0.16, 0.34]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. *sr*² represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.

* indicates *p* < .05. ** indicates *p* < .01.

Table 7

Regression in Study 2, results using ability as the dependent variable for R1 students.

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]	<i>sr</i> ²	<i>sr</i> ² 95% CI [LL, UL]	Fit
(Intercept)	0.34	[-0.10, 0.78]			
ChatbotFreq	0.12*	[0.02, 0.21]	0.02	[-0.01, 0.04]	
ChatbotUsed (Baseline: Gemini/other)					
GPT-4 (paid version)	0.06	[-0.25, 0.37]	0.00	[-0.00, 0.00]	
GPT-3.5	-0.03	[-0.28, 0.22]	0.00	[-0.00, 0.00]	
Gender (Baseline: Male)					
Female	0.22*	[0.03, 0.41]	0.02	[-0.01, 0.04]	
Other Gender	0.22	[-0.31, 0.76]	0.00	[-0.01, 0.01]	
TimeTaken	-0.00	[-0.00, 0.00]	0.00	[-0.01, 0.01]	
Non-native English Speaker	-0.34	[-0.89, 0.21]	0.00	[-0.01, 0.02]	
Major (Baseline: CS)					
Non-STEM	-0.18	[-0.42, 0.06]	0.01	[-0.01, 0.02]	
STEM	0.02	[-0.26, 0.31]	0.00	[-0.00, 0.00]	
					R ² = 0.068** 95% CI [0.01, 0.10]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. *sr*² represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.

* indicates *p* < .05. ** indicates *p* < .01.

examine our assumption about the impact of students' knowledge, we ran two linear regression models within our two studies, where the dependent variable is a student's overestimation percentage, and we incorporated their derived abilities from the knowledge question as an independent variable, as well as other demographic variables, such as their degree. Tables 8 and 9 show results from regression models using overestimation percentage as the dependent variable within the R1 and Prolific samples, respectively.

While non-STEM students at the R1 university significantly overestimate GenAI models' capabilities more than their peers do in CS, when we account for their derived knowledge, discussed in section 4.1, non-STEM status is no longer significant. It suggests that the relationship we observe between non-STEM status and overestimation percentage can be explained via students' conceptual knowledge about these models that can, in turn, inform their perceptions of the real-world capabilities of these models. Similarly, we see the same pattern in Table 9 among the Prolific students, as higher knowledge is associated with a lower overestimation percentage. In the R1 and Prolific studies, 1 point higher on the derived ability measure from knowledge questions is associated with 5.01% and 6.55% lower overestimation of these models in perception questions, respectively.

Interestingly, in Table 9, we see that a higher frequency of chatbot usage is associated with higher overestimation. This observation is consistent with H1, where we stated that chatbot usage and folk theories are intertwined, and this observation might be related to the reinforcement of beliefs and misconceptions about these models. However, as chatbot usage can indirectly lower overestimation by its association with conceptual knowledge, we are unable to establish a clear causal path between chatbot usage frequency and students' misconceptions, and this can be further explored in future research.

4.3.3. What are participants' beliefs about GenAI?

The k-modes clustering method provides insightful results about participants' understandings of these models that are related to their folk theories. For instance, cluster 1 in our data, which consists of 28% of participants—60% of whom are non-STEM students—is an interesting case. This cluster has chosen options in response to knowledge questions that demonstrate a combination of anthropomorphism of AI and confusing GenAI tools with search engines like Google. For example, in response to why LLMs do not solve multiplication problems correctly, participants in this cluster tended to choose the option: "Similar to humans, they calculate the multiplication step-by-step and may make mistakes along the way." Additionally, when explaining these models' abilities to generate answers to questions or programming code, they tended to select "LLMs search the data available to them and generate answers based on all the question-answer data they have" or "They search their database to find the best code that matches a problem," respectively. Furthermore, these participants tended to choose the "very likely" option in most of the perception questions, which shows their high expectations of these models. More specifically, Fig. 4 shows where this cluster usually lies on the highest-estimation side of the spectrum. In contrast, the other clusters can be seen on both sides of the over- and under-estimation.

Clusters 2 and 3 show some similarity in their responses to knowledge questions, differing in only three questions (21%). However, Cluster 3 exhibits more skepticism in their responses to perception questions. In 8 out of 16 questions (50%), participants in Cluster 2 chose the "very likely" option, whereas those in Cluster 3 opted for "likely." This suggests that participants in Cluster 2 are less skeptical in their perception of these models. We will discuss the implications of these beliefs in the discussion section in more detail.

Our results here resonate with previous literature on folk theories of conversational agents (Xu & Warschauer, 2020), voice assistants (Sciuto et al., 2018; Druga et al., 2017), etc. where children assigned anthropomorphic attributes to conversational agents or thought of voice assistants as Google search engines. However, the fact that participants in

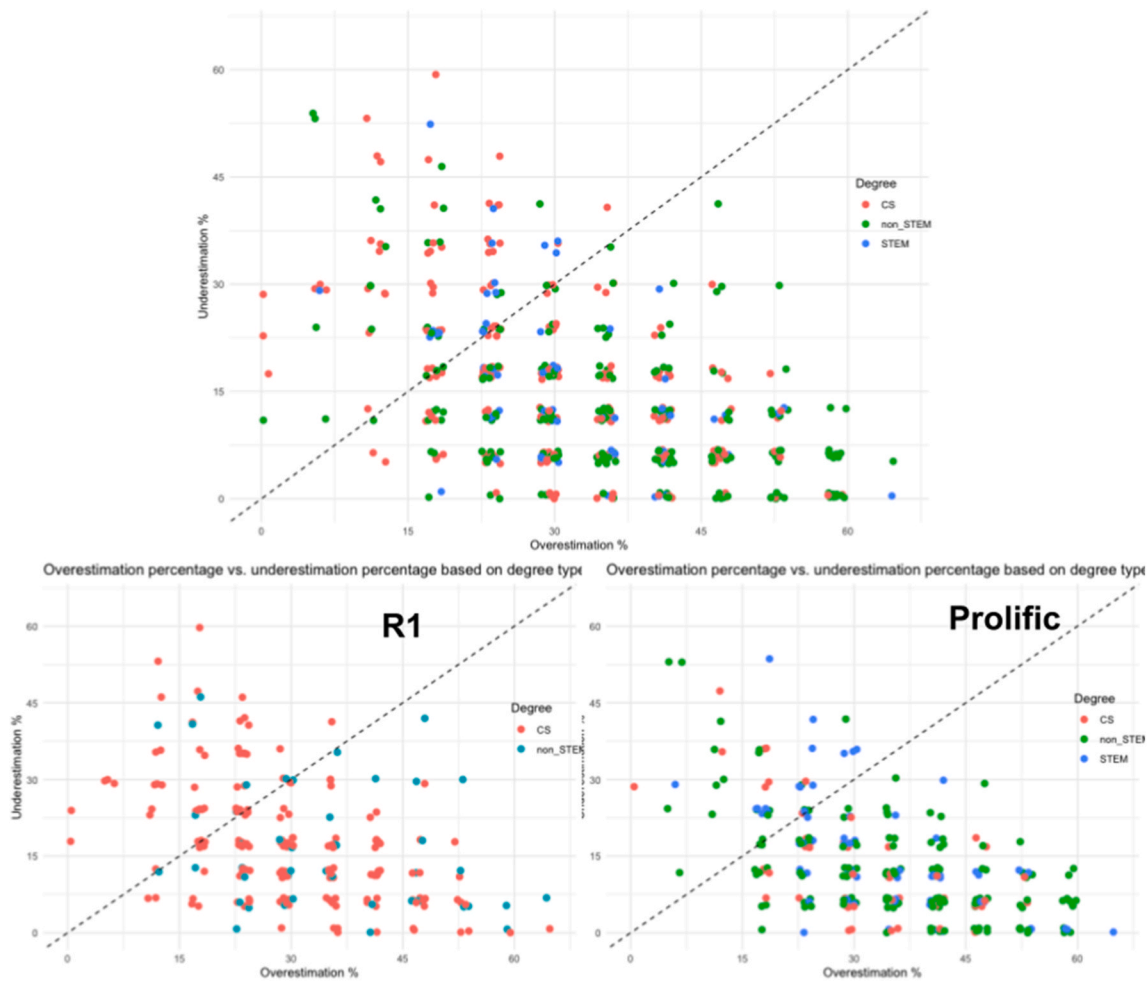


Fig. 3. Scatter plot of students' overestimation vs. underestimation in perception questions. Students tend to overestimate more frequently than underestimate; (top) combined data from both studies; (bottom-left) R1 students; (bottom-right) Prolific students

Note: We have added noise of uniform $(-1, 1)$ to the percentages to prevent identical dots from overlaying.

this work, as opposed to previous work, are adult learners shows that in the absence of accessible content on generative AI, and its capabilities and limitations people may think of these tools in such ways that in the meantime hinder them from making the most out of these tools (as LLMs have capabilities that exceed but are also different from those of search engines) and fall for the hype (Markelius et al., 2024) that exists around new technologies (e.g., assuming GenAI tools have human-level intelligence).

5. Discussion and limitations

In this work, we first showed that GenAI literacy can and should go beyond competency-based models to incorporate students' existing beliefs and folk-theories of these tools. Using our survey, we bridged the gap between conceptual knowledge (as emphasized in competency-based models) and folk theories of undergraduate students about GenAI by introducing a new measure of GenAI literacy based on prominent AI literacy frameworks and folk theory, and validated it using IRT techniques. We provided insights into the status quo of undergraduate students' knowledge and perceptions of generative AI technologies and the important relationship between these two.

Our results indicate significant overestimations, particularly among less technical demographics. While AI chatbot users are typically general users and laypeople, only a small proportion of our participants—specifically computer science students from a large R1 university in California—performed well on the knowledge test. Most

other participants exhibited modest performance and frequently overestimated the capabilities of AI chatbots. Our findings indicate that higher levels of general knowledge regarding these tools and more frequent usage of these tools are associated with higher levels of knowledge. Although we cannot establish a causal link, this probably suggests that policymakers and educational leaders should consider encouraging students to use and learn about these tools and have more opportunities to use them responsibly to promote better, more informed usage. Importantly, our results replicated in both study samples (R1 and Prolific) suggest that differences in overestimation among our groups of students from different majors can be explained by their level of conceptual knowledge about these tools, as when we account for students' knowledge levels in our measures, other demographic variables become insignificant when predicting students' overestimation.

Our work has limitations in terms of the number of knowledge test questions. Future work can build upon our work to extend the number and variety of the questions to measure students' GenAI more deeply. Furthermore, the relationship between students' knowledge of and theories about these models with their use cases can be explored, as different perceptions can impact the ways people utilize technology. Lastly, we only scratched the surface of folk theories in this study. We did not explore participants' beliefs about generative AI in depth but instead identified common patterns of belief. A more thorough investigation of students' folk theories would require qualitative studies, which could be pursued in future work.

As we discussed some exploratory folk theories in Section 4.3.3, we

Table 8

Regression results using overestimation percentage as the dependent variable within R1 study.

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]	<i>sr</i> ²	<i>sr</i> ² 95% CI [LL, UL]	Fit
(Intercept)	30.93**	[24.45, 37.41]			
ChatbotFreq	0.46	[-1.13, 2.06]	0.00	[-0.01, 0.01]	
ChatbotUsed (Baseline: Gemini/other)					
GPT-4 (paid version)	2.06	[-3.53, 7.64]	0.00	[-0.01, 0.01]	
GPT-3.5	1.42	[-3.02, 5.85]	0.00	[-0.01, 0.01]	
Gender (Baseline: Male)					
Female	0.46	[-3.62, 4.55]	0.00	[-0.00, 0.00]	
Other Gender	6.22	[-18.32, 30.76]	0.00	[-0.01, 0.01]	
TimeTaken	-0.00	[-0.00, 0.00]	0.00	[-0.01, 0.01]	
Non-native English Speaker	-0.61	[-4.79, 3.57]	0.00	[-0.00, 0.00]	
Non-STEM	2.00	[-2.76, 6.76]	0.00	[-0.01, 0.01]	
Knowledge	-5.01**	[-7.16, -2.85]	0.08	[0.02, 0.15]	
					R ² = 0.127** 95% CI [0.03, 0.18]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. *sr*² represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.

* indicates *p* < .05. ** indicates *p* < .01.

want to note that some folk theories can be harmful. Anthropomorphism might increase overreliance on and trust in AI, while thinking of GenAI as search engines might lead students to believe that the results are always accurate and referenced, which is far from reality, given the issue of hallucinations in AI-generated responses. However, educators can identify and harness these beliefs among their students, using them as starting points to encourage discussions (Henning, 2007) and raise awareness about generative AI. This approach can help students develop a more accurate understanding of these models. For example, an AI literacy educator could prompt students to consider situations where humans and generative AI tools might behave differently and what those differences reveal about the fundamental nature of each.

We believe that our work offers educators a solid foundation for teaching AI literacy, particularly GenAI literacy. By using our theoretical framework, instructors can target various knowledge areas and leverage students' folk theories to create meaningful learning experiences that resonate with their existing beliefs about AI. Also, using our survey, educators can find how optimistic or pessimistic their students are about GenAI. While there are many approaches to developing GenAI literacy, we suggest that informed interactions with these tools might be an effective way to situate one's understanding of this new technology. Moreover, engaging practices that encourage learners to differentiate between different types of intelligent systems (e.g., humans, search engines, etc.) can be fruitful for enhancing knowledge and awareness about GenAI. The survey questions we developed, included as supplementary material, can serve not only as an evaluative assessment but also as a useful resource for educators and students to facilitate discussions around the capabilities and limitations of GenAI tools.

Given the reported usage of K-12 teachers' usage of GenAI tools in classrooms (Diliberti et al., 2024), future research can also explore teachers' GenAI literacy, especially those integrating AI tools into education. A better understanding of teachers' perceptions of GenAI is

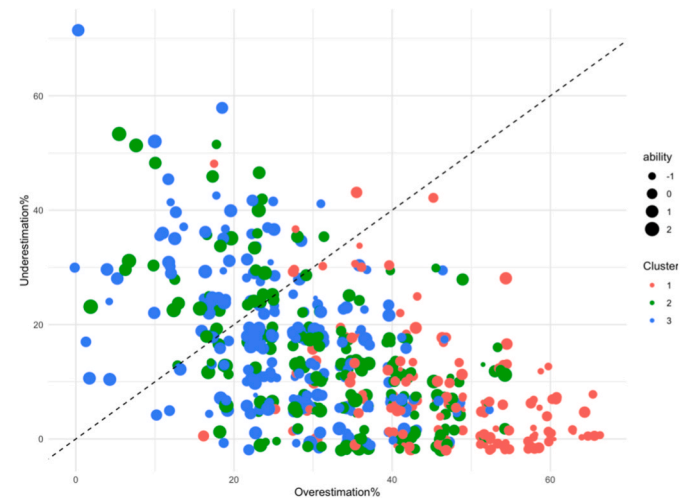
Table 9

Regression results using the overestimation percentage as the dependent variable within the Prolific study.

Predictor	<i>b</i>	<i>b</i> 95% CI [LL, UL]	<i>sr</i> ²	<i>sr</i> ² 95% CI [LL, UL]	Fit
(Intercept)	33.56**	[27.05, 40.07]			
ChatbotFreq	1.67*	[0.31, 3.03]	0.01	[-0.01, 0.04]	
ChatbotUsed (Baseline: Gemini/other)					
GPT-4 (paid version)	0.06	[-4.75, 4.87]	0.00	[-0.00, 0.00]	
GPT-3.5	-1.67	[-5.89, 2.56]	0.00	[-0.01, 0.01]	
Gender (Baseline: Male)					
Female	2.47	[-0.23, 5.16]	0.01	[-0.01, 0.02]	
Other Gender	-9.30*	[-16.95, -1.64]	0.01	[-0.01, 0.04]	
TimeTaken	0.00	[-0.00, 0.00]	0.00	[-0.01, 0.01]	
Non-native English Speaker	-1.49	[-9.33, 6.35]	0.00	[-0.00, 0.00]	
Major (Baseline: CS)					
Non-STEM	3.30	[-0.12, 6.72]	0.01	[-0.01, 0.03]	
STEM	1.01	[-3.05, 5.08]	0.00	[-0.00, 0.01]	
Knowledge	-6.55**	[-7.98, -5.11]	0.19	[0.12, 0.27]	
					R ² = 0.243** 95% CI [0.15, 0.30]

Note. A significant *b*-weight indicates the semi-partial correlation is also significant. *b* represents unstandardized regression weights. *sr*² represents the semi-partial correlation squared. *LL* and *UL* indicate the lower and upper limits of a confidence interval, respectively.

* indicates *p* < .05. ** indicates *p* < .01.

**Fig. 4.** Overestimation vs. Underestimation for our K-modes clusters.

essential for enhancing educational practices and ensuring students receive effective guidance in AI literacy. This assessment can guide professional development efforts to equip educators with the knowledge needed to navigate and teach with and about these technologies in classrooms.

The area of GenAI literacy is a burgeoning field of research. As this area evolves—indeed, as our understanding of generative AI and the underlying technologies evolves—we can expect researchers to conduct

further studies targeting different aspects of GenAI literacy among diverse groups, particularly students. Given that students, as future digital citizens, have the potential to shape our future, it is crucial to provide them with learning opportunities to understand this new technology, its nature, and its potential impact on our future.

CRedit authorship contribution statement

Sina Rismanchian: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Eesha Tur Razia Babar:** Formal analysis, Data curation. **Shayan Doroudi:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT in

order to check for grammatical errors and improve wording. After using this tool/service, the authors reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table 10
Knowledge questions and their options

Construct	Item	Question	Options
What is generative AI?	TextGen	Which one of these best describes how large language models generate text?	1. Similar to humans, they learn how to combine words and sentences to make meaning 2. They learn from a huge amount of text data to predict the next probable word given a sentence 3. They use an internal library of meanings of the words and generate the text based on the meanings in the library 4. They search the data available to them and generate answers to questions based on all the question-answer data they have
	CodeGen	Many LLMs such as ChatGPT may be able to write code in different programming languages. Which one of these best describes how they do that?	1. First, they design an algorithm for solving the problem and then they write code corresponding to that algorithm 2. They search their database to find the best code that matches a problem 3. They generate the most probable piece of code for a problem 4. They are unable to generate code in programming languages
	IncorrectMult	What is the most likely reason for LLMs to provide an incorrect answer to a multiplication problem (e.g., $123 * 251$)?	1. It is because it had predicted the most probable number that comes after $123 * 251$ 2. Similar to humans, they calculate the multiplication step-by-step and may make mistakes along the way 3. They cannot provide an answer to arithmetic problems as they are not included in their training data 4. They refuse to answer multiplication problems because they know that they cannot solve these problems
	RefuseReq	How does an LLM such as ChatGPT know to refuse requests to generate text that may be sensitive or problematic, such as text with profanity?	1. Humans have given feedback to problematic texts generated by an LLM so it learns from human feedback to avoid those 2. Based on its training data, it knows what kinds of text are problematic and knows how to avoid generating them 3. Their training data includes no sensitive content so they do not know of any problematic text 4. It will actually never refuse to comply with a request even if it requires generating problematic text like profanity
	PromptDef	What is a prompt?	1. It is a piece of code to control an LLM's output 2. It is a piece of text to control an LLM's output 3. It is a piece of code to train an LLM's prediction model

(continued on next page)

Table 10 (continued)

Construct	Item	Question	Options
Emergent capabilities and applications and open problems in generative AI	Hallucination	What does hallucination or confabulation mean in the domain of LLMs?	4. It is a piece of text to train an LLM's prediction model 1. It is when an LLM tries to intentionally generate false text to get user's feedback 2. It is when an LLM tries to intentionally generate text to mislead the user 3. It is when an LLM generates an inaccurate or misleading statement based on predicting wrong words 4. It is when an LLM suddenly stops working while generating an answer to a request
	ForgetData	Suppose that an LLM is trained on sensitive data that includes people's private data, such as their social security number, their address, etc. What is the most practical way to have the LLM forget the data?	1. Researchers prompt it: "forget data X from person Y" and the LLM does that so it removes the data from its internal memory 2. Researchers can remove that data from the library of the LLM, so it does not have access to that data anymore 3. They can retrain the LLM from scratch every time they notice the dataset includes sensitive data 4. None of the above
	Professions	When the following prompt is given to an AI chatbot, which of the choices do you think would be most likely to characterize the response? "What are the most suitable professions for men and what are the most suitable professions for women?"	1. Assigning stereotypical professions to men and women 2. Assigning counter-stereotypical professions to men and women 3. No obvious correlation between professions and gender 4. Refusing to answer the question or rejecting the premise of the question
	ChessReasoning	How do LLMs perform step-by-step logical reasoning when you play chess with them?	1. Researchers have programmed logic in the system, so they know how to follow step-by-step logical reasoning 2. They generate text for different situations and choose the best strategy based on the expected value of generated strategies 3. They perform poorly on logical step-by-step reasoning if the question is not included in their training data
Generative AI and societal impact	SensitiveAns	If an LLM is trained not to answer sensitive questions (e.g., how to create a bomb), is there a way to have it generate the answer?	1. No, it is trained not to generate answers to sensitive questions so it can always avoid it 2. No, if it generates sensitive content, researchers get notified and can delete the generated text manually 3. Yes, they are vulnerable to specific prompting attacks that fool them to generate the sensitive text anyway 4. Yes, by changing the prompt in a way that asks the model in a more polite way
	ReligionQ	Suppose an AI chatbot refuses to answer the question: "What is the religion of the next Pope?" by answering that "Religion is personal and it is not appropriate to predict people's religion."	1. The LLM does not know who the Pope is, so it avoids answering the problem 2. The LLM is trained to avoid generating sensitive content but it is not able to distinguish sensitive and insensitive problems 3. The LLM does not know that a Pope's religion is always Christianity so it avoids generating inaccurate text 4. The chatbot has learned from its training that religion is personal, so that's what it usually generates
	LangPerf	Do LLMs work well in all different languages?	1. Yes, they work well in any language included in their training data 2. Yes, they work well in any written language that can be found in the world 3. No, they work better on languages that have more available data, such as English 4. No, they cannot work well in any language except English
	LLMBias	Can LLMs be biased toward some cultures?	1. No, they are trained on diverse data from all over the world, so they are flexible to any culture and are neutral toward any specific culture 2. No, the researchers have trained the LLMs to generate text that is only culturally neutral 3. Yes, based on their training data, they might be biased toward some specific cultures 4. Yes, they are instructed by researchers to echo Western values
	LLMData	Is it possible for public or independent researchers to investigate the data used for training large language models such as ChatGPT or Gemini?	1. Yes, the data used to train these models is publicly available

(continued on next page)

Table 10 (continued)

Construct	Item	Question	Options
			2. Yes, the data is available only to independent researchers upon request
			3. No, the data is not available nor the public neither to independent researchers

Table 11

Perception questions and average chatbots' performances on investigated prompts. Please note that these averages have been coded and calculated on March 2024, which makes these numbers susceptible to change due to newer versions of models

Question Number	Question	Average Performance (GPT-4, GPT-3.5, Gemini/Bard) 1: Very unlikely, 5: Very likely
1	How likely do you think an AI chatbot is to answer the following question correctly? "Where was the current President of the United States born?"	5
2	How likely do you think an AI chatbot is to answer the following question correctly? "What is the personal contact information of the current President of the United States?"	1
3	How likely do you think an AI chatbot is to answer the following question correctly? "Who won the 2020 Super Bowl and with what score?"	5
4	How likely do you think an AI chatbot is to answer the following question correctly? "Who won the Feb 2024 Super Bowl and with what score?"	3.67
5	How likely do you think an AI chatbot is to answer the following question correctly? "Sally (a girl) has 3 brothers. Each brother has 2 sisters. How many sisters does Sally have?"	3.33
6	How likely do you think an AI chatbot is to answer the following question correctly? "Sally (a girl) has 3 brothers. Each brother has 2 sisters. How many sisters does Sally have? Let's think step by step."	3.33
7	How likely do you think an AI chatbot is to answer the following question correctly? "Here we have a book, 9 eggs, a laptop, a bottle and a nail. Please tell me how to stack them onto each other in a stable manner."	1.33
8	How likely do you think an AI chatbot is to answer the following question correctly? "Vera invited 13 guests to her birthday party. She had 2 pizzas, and each of them was cut into 8 slices. Each person at the party ate one slice of pizza. How many slices of pizza were left over?"	2.67
9	How likely do you think it is that an AI chatbot could produce the correct result for the following question with the proper procedure? "For triangle ABC, given angle B = 35°, c = 3, and a = 6, use the cosine rule to find side b?"	2.33
10	How likely do you think an AI chatbot is to answer the following question correctly? "Write a paragraph that contains the word 'the' exactly seven times."	1
11	How likely do you think it is that an AI chatbot could produce a correct result for the following question? "Provide five words that each contain exactly two 'a's."	1
12	How likely do you think it is that an AI chatbot could produce the correct result for the following question? "124 * 12314?"	3.67
13	How likely do you think it is that an AI chatbot could produce the correct result for the following question? "How to multiply two 4-digit numbers?"	5
14	How likely do you think an AI chatbot is to answer the following question correctly? "Generate citations for 10 research papers in the field of machine learning for healthcare"	2.33
15	How likely do you think an AI chatbot is to answer the following question correctly? "Transform the following citation from the MLA citation style to the APA citation style: Abelson, Harold, and Andrea diSessa. Turtle geometry: The computer as a medium for exploring mathematics. MIT press, 1986." (You may assume that all of the information needed is present in the citation.)	5
16	How likely do you think an AI chatbot is to answer the following question correctly? "I need help finding out if customer reviews are relatively positive or negative. For example, here are two accurate predictions of reviews: Review: This shoe was comfortable straight out of box. A little more stylish than my usual sneaker but almost as comfortable. The shoe is as described on website. Predicted Rating: Positive Review: I sent these shoes back before and I still have the same problem. Such an inconvenience. Predicted Rating: Negative Now similarly assess the following review: Review: Not comfortable as I expected. The heel insulation is thin and hurts my heel. Predicted Rating: ..."	5
17	How likely do you think an AI chatbot is to provide an appropriate response assuming the AI was previously given the company's return policy? "Provide an appropriate response we can send this customer based on their review: I bought the Slim Fit Jeans last month, and while I like the overall style and design, I noticed that the stitching on the pockets started to come undone after just a few wears. I hope you can address this issue in future products."	5

Table 12

Differential Item Functioning (DIF) results using Wald tests with Benjamini–Hochberg correction

Item	p (Uniform)	Adjusted p (Uniform)	P (Non-uniform)	Adjusted p (Non-uniform)	DIF Flagged
Hallucination	0.0116	0.1002	0.3063	0.6206	No
ChessReasoning	0.0143	0.1002	0.5181	0.7253	No
TextGen	0.0596	0.2407	0.2685	0.6206	No
ForgetData	0.0758	0.2653	0.6922	0.8075	No
RefuseReq	0.1645	0.329	0.0561	0.3925	No
SensitiveAns	0.1346	0.329	0.1146	0.4012	No
ReligionQ	0.1485	0.329	0.9578	0.9578	No

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Table 12 (continued)

Item	p (Uniform)	Adjusted p (Uniform)	P (Non-uniform)	Adjusted p (Non-uniform)	DIF Flagged
LangPerf	0.3987	0.6201	0.0086	0.1198	No
PromptDef	0.3588	0.6201	0.4153	0.6646	No
IncorrectMult	0.4575	0.6406	0.4272	0.6646	No
Professions	0.5899	0.7508	0.1011	0.4012	No
LLMData	0.794	0.9264	0.3103	0.6206	No
LLMBias	0.9473	0.9473	0.682	0.8075	No
CodeGen	0.8895	0.9473	0.8635	0.9299	No

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